

RoboRacer Modeling, Controls, and LiDAR Mast Design Review

Item 12 review-ready report - 2026-07-24

Decision state: the simulation/controls evidence and LiDAR-mast analysis are complete. The mast analysis caught a modal failure, drove a geometry revision, and validated the revised design with converged finite-element results. CAD, fabrication, as-built inspection, and physical compliance are downstream gates.

Physical evidence state: Experiment registered - compliance campaign scheduled for July 25-August 2, 2026; measurement pending. The protocol and verdict thresholds are frozen. No physical result is claimed in this report.

Scope guardrail: controller, estimator, and system-identification results come from simulation. Mast strength, deflection, and frequency results are hand calculations and FEA, not measurements. The planned compliance experiment observes displacement per force and **does not validate stress**.

Primary reproduction: `./run_all.sh` for the simulation pipeline; Section 12 gives the mechanical and report commands.

Executive Review

What was decided

The project has two connected evidence chains:

```
Gym experiment
-> timestamped CSV telemetry
-> numerical/model validation
-> held-out identification
-> controller, estimator, and robustness decisions

Committed simulation envelope + locked sensor mass
-> hand calculation
-> 174.7 Hz modal failure against a 200 Hz guard
-> shorter/larger mast redesign
-> 330.1 Hz hand prediction
-> 285.5 Hz converged FEA result
-> physical compliance gate (registered; measurement pending)
```

The baseline 120 mm-long, 16 mm-OD, 1.5 mm-wall 6061-T6 tube passed the strength check but failed its predeclared frequency guard: **174.7 Hz < 200 Hz**. A sweep selected a 100 mm-long, 20 mm-OD tube with the same material and wall. The revised hand model predicted **330.1 Hz** and the higher-fidelity FEA predicted **285.5 Hz**, which clears the guard by 1.43x. The final mesh change was **-0.15% gauge stress, +0.30% tip deflection, and -1.42% first frequency**, all inside the 5% convergence criterion.

Static FEA also agreed with the hand model inside the frozen +/-15% credibility band: **17.4 MPa versus 17.1 MPa** at the mid-span gauge (+1.3%), **0.176 mm versus 0.166 mm** tip deflection (+5.9%), and **285.5 Hz versus 330.1 Hz** first frequency (-13.5%). These are model-to-model checks. The registered physical campaign will judge measured compliance separately under the same +/-15% rule.

Why the failure matters

The mast story is not "FEA confirms a chosen shape." The analysis rejected the first design despite ample yield margin, identified stiffness as the binding constraint, and spent two direct design levers - length and diameter - to fix it. That decision cost only 1.4 g of mast mass while improving the crash-case yield safety factor from 4.05 to 8.05.

1. Scope, Motivation, and Evidence Labels

The practical motivation is to build credible RoboRacer engineering evidence before access to a complete physical vehicle. The simulation lane establishes repeatable modeling, identification, control, estimation, and failure-analysis methods. The parallel design lane specifies a restricted-class hardware platform and applies those simulation-derived loads to a buildable mechanical decision. This is the simulation/control and design-review half of the portfolio; future hardware data must earn its own claims.

Evidence labels used in this report

Label	Meaning	Examples
SIMULATION OUTPUT	Generated by a committed experiment or run	lateral acceleration, lap metrics, controller timing
TESTED SOFTWARE	Executable gate covered by regression tests	ROS-bag evidence checks, physical-verdict analyzer
HAND CALC	Closed-form engineering model with stated assumptions	mast stress, deflection, Rayleigh frequency
FEA	gmsh + CalculiX result on stated geometry/BCs	gauge stress, tip deflection, modes
DATASHEET / CATALOG	Selected-part or rule value	LiDAR mass, chassis mass, supply limits
ASSUMED	Engineering input awaiting CAD, certificate, or inspection	shock level, bracket mass, interface tolerances
PHYSICAL PENDING	Frozen test exists; accepted data does not	mast compliance campaign

No SIMULATION OUTPUT or FEA value is described as a physical measurement. Catalog values and assumed allowances remain visibly distinct in the mass, load, and tolerance budgets.

Report boundary

This report closes the item-12 design review. It does not close:

- parametric mast/deck CAD or per-component CG coordinates;
- the fabrication traveler and as-built dimensional inspection;
- the calibrated compliance campaign;
- physical vehicle system identification; or
- on-car vibration, perception, fatigue, or crash qualification.

Those are observable downstream gates rather than gaps hidden inside the report.

2. Requirements and Verification

The matrix below freezes the main review claims and the evidence that resolves each one. The full architecture is in docs/design/13_requirements_architecture.md.

ID	Requirement / criterion	Evidence and result	State
R-05	Size actuation and structure against a traceable dynamics envelope	Clean selected lap reaches 19.436 m/s ² peak lateral and 9.51 m/s ² peak longitudinal acceleration (runs/ride_quality_baseline/summary.json)	SIMULATION OUTPUT
R-08	100 Hz compute with p95 <= 25% of 10 ms	MPC p95 = 1.32644 ms (runs/mpc_controller/results.csv)	SIMULATION OUTPUT
R-11	Ready-to-race mass <= 4.5 kg	Bottom-up locked-part estimate is 4.42 kg with 0.08 kg margin; assumed rows remain flagged (docs/design/16_mechanical_design_analysis.md)	CATALOG + ASSUMED
R-12	100 Hz control loop, zero-order held	Shared controller runs use 100 Hz commands and RK4 integration	SIMULATION OUTPUT
M-01	First mast mode >= 200 Hz	Baseline 174.7 Hz FAIL; revised FEA 285.5 Hz PASS	HAND CALC + FEA
M-02	Crash yield safety factor >= 1.5	Revised hand result 34.3 MPa and SF 8.05	HAND CALC
M-03	Hand/FEA headline agreement within +/-15%	Frequency -13.5%, deflection +5.9%, gauge stress +1.3%	HAND CALC + FEA
M-04	Final mesh change < 5% on reportable metrics	-0.15%, +0.30%, -1.42%; PASS	FEA
M-05	Scan plane clears 0.30 m wall at 10 m	Blind 1.284 deg FAIL; leveled 0.354 deg PASS; deck <= 0.138 m	HAND CALC + ASSUMED
M-06	As-built compliance agrees with as-built FEA within +/-15% after quality gates	Experiment registered; campaign scheduled; measurement pending	PHYSICAL PENDING

Architecture implications

The selected Traxxas Slash 4x4 platform has a 0.324 m wheelbase rather than the 0.3302 m simulation value, and the bottom-up 4.42 kg design estimate exceeds the 3.74 kg simulation model by about 18%. The real tires also differ. Therefore the identified simulator coefficients are not transferred to hardware. The hardware path repeats excitation, fitting, and held-out validation.

The requirements matrix makes the division explicit: simulation sizes and screens the design; catalog/assumed inputs support the current mechanical model; as-built inspection and physical data replace them later.

3. Modeling Pipeline and Numerical Credibility

The simulator pipeline uses one importable gym/robo racer/ code path for numerics, dynamics, telemetry I/O, tracks, pure pursuit, estimation helpers, and closed-loop runs. That reduces the risk that separate scripts silently use different equations or conventions.

Timestep selection

The RK4 sweep declared its criterion before interpreting results:

- RMS progress-aligned change between refinements < 0.05 m;
- maximum progress-aligned change < 0.2 m.

The selected $\text{dt} = 0.0020$ s is the coarsest tested timestep that clears both: the next-finer comparison gives **0.0133307 m RMS** and **0.176077 m maximum** progress-aligned change. The original 0.01 s baseline misses the maximum-change criterion at 0.277026 m. Source: [reports/integrator_convergence.md](#).

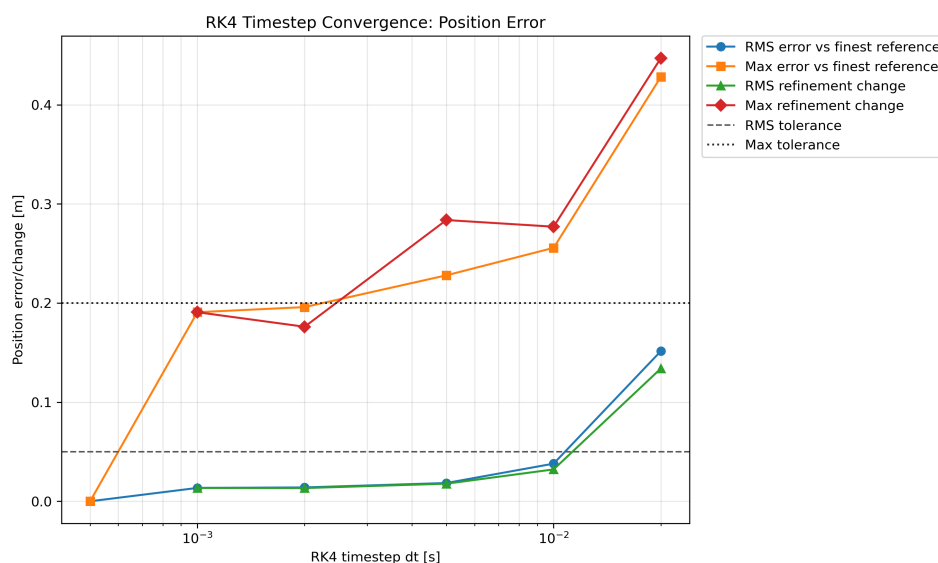


Figure 1. RK4 position-error convergence across the tested timestep refinements.

Kinematic-to-dynamic boundary

The kinematic replay accumulates 66.7991 m position RMSE across the full dynamic lap, while its instantaneous yaw law matches Gym at low speed with a median ratio of 0.999. The combination rules out a simple sign, wheelbase, or timestep bug and points to the expected missing lateral/yaw dynamics. Figure 2 shows the resulting trajectory error. Source: [reports/model_vs_gym_comparison.md](#).

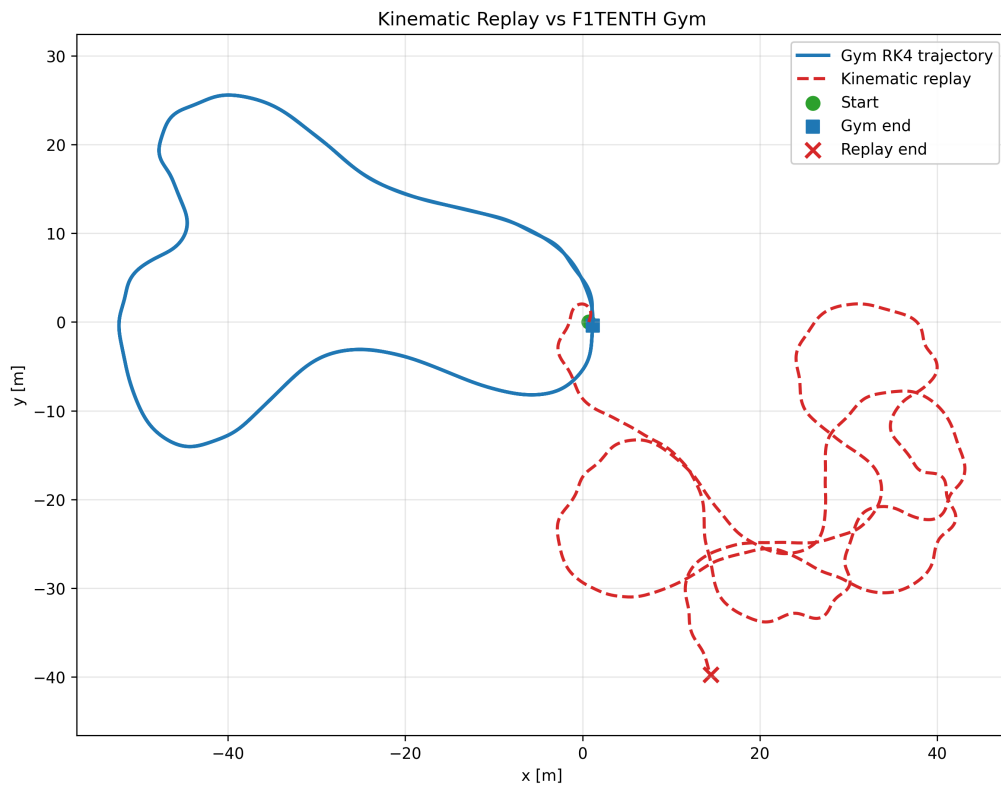


Figure 2. Kinematic replay versus Gym trajectory; the growing error motivates the nonlinear single-track model.

4. Identification and Held-Out Validation

The nonlinear single-track coefficients C_{Sf} and C_{Sr} are fit with bounded nonlinear least squares on normalized one-step yaw-rate and slip-angle residuals. Intervals below 0.75 m/s are excluded because Gym uses a low-speed kinematic fallback there. The first 70% of eligible intervals form the fit set; the final 30% are held out chronologically. The held-out rollout begins from one measured split state and then propagates recursively without state resets.

Parameter	Identified	Gym oracle	Relative error
C_{Sf}	4.717999998	4.718000000	5.197e-10
C_{Sr}	5.456200004	5.456200000	6.527e-10

The independent held-out replay passes all declared gates:

- yaw-rate VAF 100.000%, RMSE 3.028881e-10 rad/s;
- slip-angle VAF 100.000%, RMSE 2.877898e-10 rad;
- yaw RMSE 3.846e-10 rad;
- position RMSE 2.067e-09 m;
- normalized-residual Jacobian condition number 1.340140; and
- C_{Sf}/C_{Sr} parameter correlation 0.090721.

Source: [reports/dynamic_parameter_identification.md](#) and [runs/parameter_id_robustness/metrics.csv](#).

Interpretation

This is an oracle-recovery check: clean data was generated by the same simulator model being fitted. The result proves that the pipeline, parameterization, and held-out gate are internally coherent. It does not prove that these coefficients describe a physical car. Section 6 deliberately degrades the data to show where that credibility fails.

5. Controller and Estimator Decisions

Pure pursuit, LQR, and constrained MPC use the same map, RK4 $dt = 0.002$ s, and 100 Hz zero-order-held command. LQR and MPC apply bounded corrections on top of the selected pure-pursuit feedforward so the comparison asks whether model-based correction improves the same baseline.

Controller	Lap time (s)	RMS CTE (m)	Max CTE (m)	Steering effort (rad)
Pure pursuit	38.042	0.157892	0.403062	8.19853
LQR	38.064	0.178628	0.497708	8.98777
MPC	37.860	0.169238	0.476650	8.67710

Decision: pure pursuit remains the tracking baseline because it has the lowest RMS CTE and steering effort. LQR is stable and survives the defined offset and delay cases, but does not improve nominal tracking. MPC earns its place through explicit steering/rate/acceleration constraints and a p95 solve time of **1.32644 ms**, not lower CTE. Its 36.807 ms maximum solve-time spike exceeds one 10 ms control period and remains a deployment risk.

Controller sources: runs/controller_comparison/results.csv and runs/mpc_controller/results.csv.

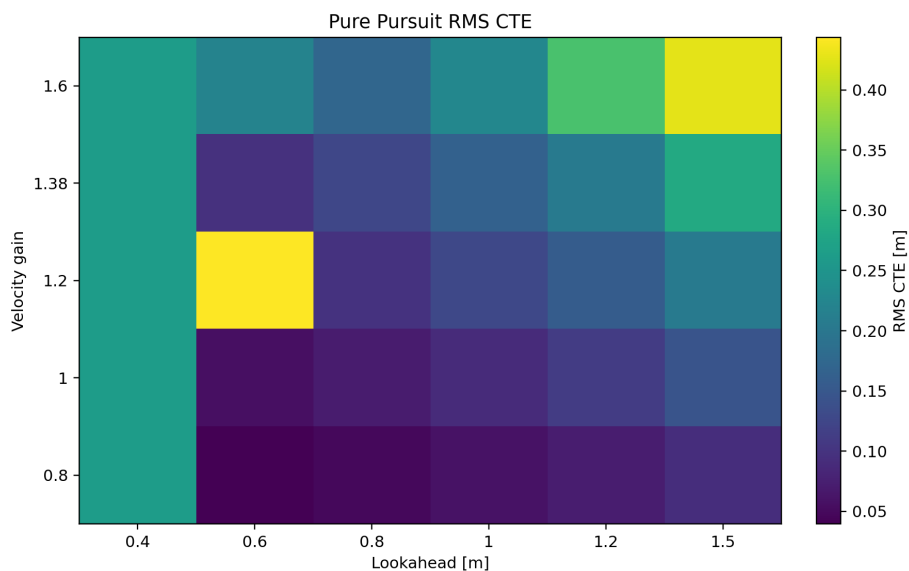


Figure 3. Pure-pursuit sweep used to select the shared tracking baseline.

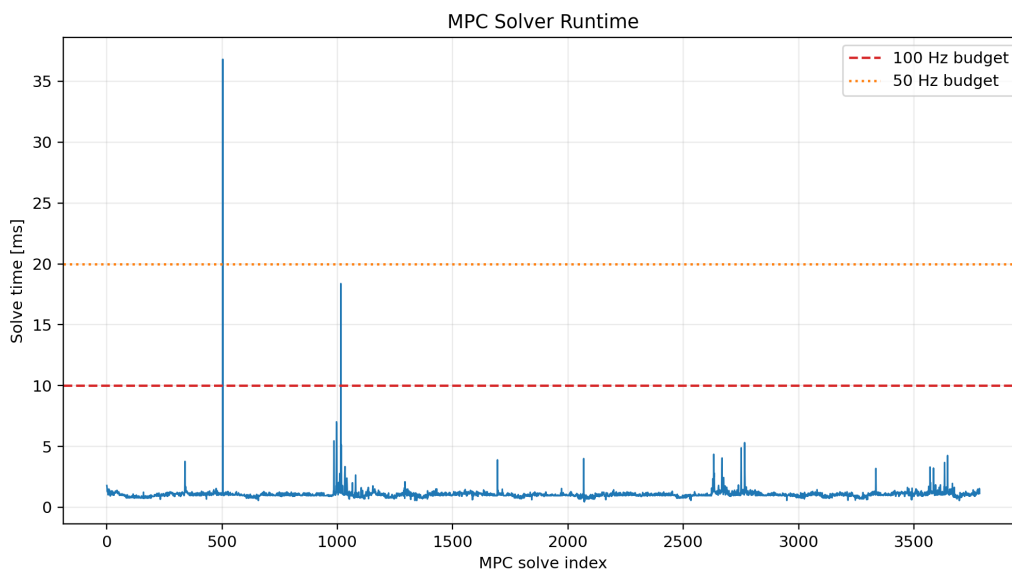


Figure 4. MPC timing distribution against the real-time control budget.

State estimation

The EKF is tested under clean, noisy, and dropout scenarios against deliberately mis-initialized dead reckoning. Dead reckoning reaches 46.10 m position RMSE. The EKF stays below 1 m RMSE even in the 3 s dropout case (0.705808 m), though the transient peak reaches 5.07754 m.

Estimator source: `runs/ekf_study/summary.csv`.

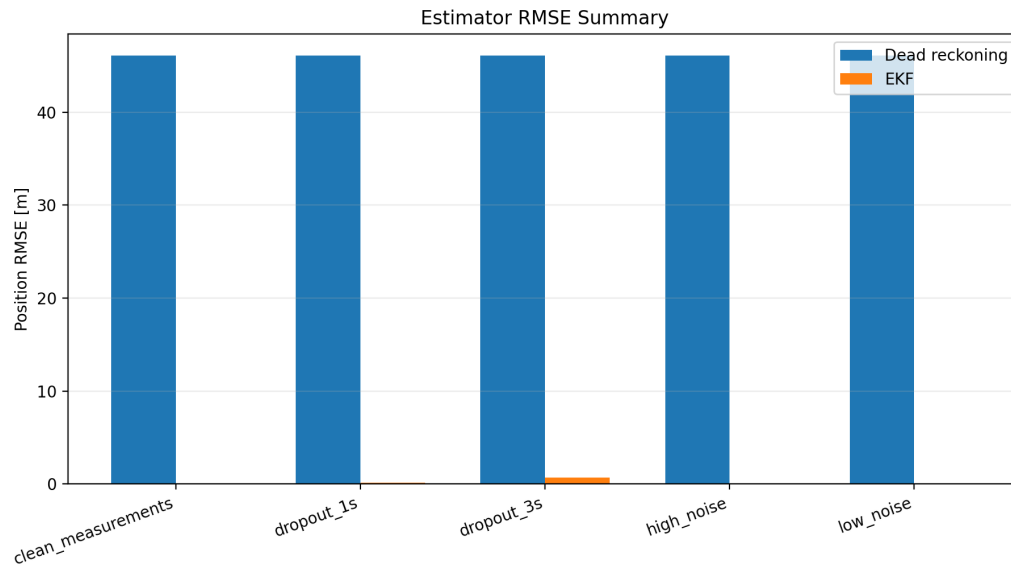


Figure 5. EKF position RMSE under noise and dropout scenarios.

6. Failure Modes and Sim-to-Real Robustness

The FMEA reproduces or quantifies seven controller, estimator, numerical, latency, and actuator cases. The highest RPNs are:

Scenario	Detection signal	RPN	Decision
100 ms latency	collision at 1.484 s	200	timestamp, reduce delay, compensate
high sensor noise	EKF RMSE 0.053 m	180	tune covariance, reject outliers
3 s dropout	EKF RMSE 0.706 m	168	detect stale data, slow or stop

High noise and the Euler case are quantified but not driven to a hard failure, so their reproduced flags remain false. That distinction is preserved rather than converted into a stronger claim.

Identification robustness

The near-oracle parameter fit first fails acceptance at noise_medium on the held-out yaw-rate gate. Latency is more damaging: 20 ms drives the C_Sf relative error to approximately 0.440, and 50 ms drives it to approximately 0.892. The hardware identification plan therefore treats timestamp integrity and end-to-end latency as first-order acceptance concerns.

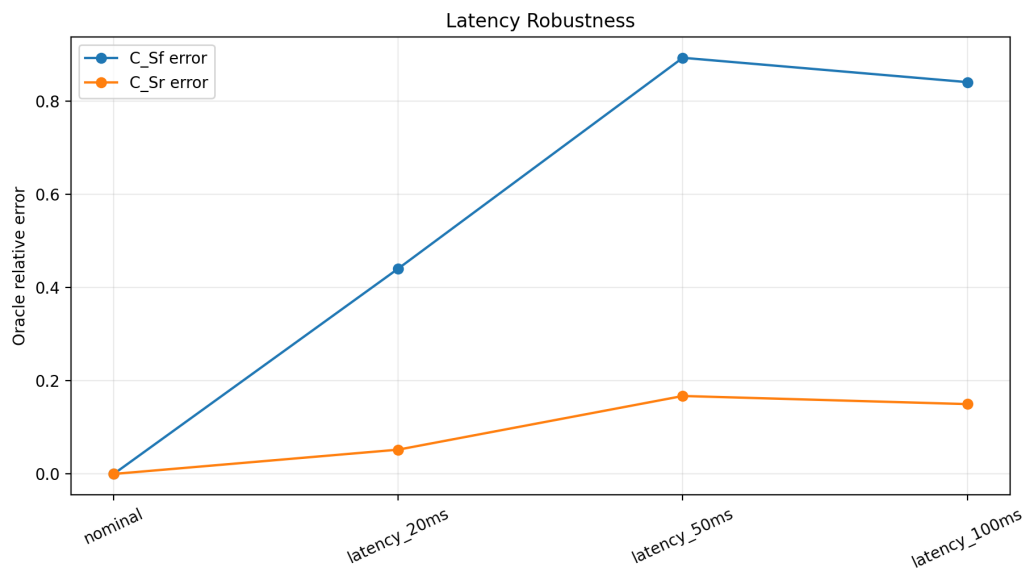


Figure 6. Parameter-identification degradation under injected input latency.

Sources: [reports/failure_mode_fmea.md](#), [reports/parameter_id_robustness.md](#), [runs/failure_mode_fmea/results.csv](#), and [runs/parameter_id_robustness/metrics.csv](#).

7. Vehicle Architecture, BOM, and Mass Budget

The hardware design uses the named restricted-class Traxxas Slash 4x4 VXL baseline, a VESC 6 MkVI, the stock Velineon 3500 motor and 2075 servo, a Jetson Orin NX 16GB on a Saeed J401 carrier, a Hokuyo UST-10LX LiDAR, VESC IMU/ERPM odometry, and separate 3S drive/compute batteries.

Key interfaces stay aligned with the software pipeline:

```
Hokuyo UST-10LX -> /scan -> localization -> /ego_racecar/odom
VESC IMU + ERPM -> sensor fusion / odometry -> /ego_racecar/odom
controller -> /drive -> VESC + steering servo
```

Mass budget

The committed bottom-up table totals **approximately 4.42 kg** against the R-11 **4.5 kg** ceiling. The 0.08 kg margin is thin and several rows are explicit allowances:

- 2.640 kg rolling chassis before removals/additions;
- 0.175 kg mast tip assembly, including 0.130 kg LiDAR;
- 0.376 kg drive battery and 0.376 kg compute battery;
- catalog masses where available; and
- ASSUMED deck, bracket, fastener, harness, and removal allowances.

This is a design estimate, not a scale measurement. It triggers two downstream checks: extract CG coordinates/mass properties from CAD, then replace every allowance with as-built scale data. The simulation coefficients are not carried across the resulting mass, geometry, and tire deviations.

Sources: [docs/design/14_chassis_drivetrain_actuators.md](#), [docs/design/15_sensor_compute_power.md](#), and [docs/design/16_mechanical_design_analysis.md](#).

8. Mechanical Design and Analysis

Load path and assumptions

The mast is modeled as a fixed-root cantilever carrying a 0.175 kg tip assembly. The mass consists of the 0.130 kg Hokuyo body plus assumed bracket, fastener, and cable allowances. The clean selected simulation lap supplies **19.436 m/s²** peak lateral acceleration. A 50 g shock level with 1.5 safety factor is analyzed separately and governs strength.

```

LiDAR + bracket + cable, m_tip = 0.175 kg
    |
    v lateral tip load
  [100 mm mast]
    |
fixed deck interface
    |
chassis / tires
  
```

The baseline hand calculation is committed at `runs/mast_hand_calc/summary.txt`. It uses a 120 mm-long, 16 mm-OD, 1.5 mm-wall 6061-T6 tube. The maneuver load is 6.79 N. The 50 g shock case applies **128.76 N**, producing 68.10 MPa root bending stress, 0.5931 mm tip deflection, and yield safety factor 4.05. Strength passes.

The binding failure

The same baseline produces **174.7 Hz** in the Rayleigh tip-mass model. The criterion requires the first mode to clear the 100 Hz control rate and a low-hundreds-Hz drivetrain band with a **200 Hz guard**. The baseline therefore fails even though its strength margin is comfortable.

Redesign

The committed sweep at `runs/mast_hand_calc/design_sweep.txt` evaluates length, diameter, wall, and material. The selected configuration is:

Attribute	Baseline	Revised
Material	6061-T6	6061-T6
Length	120 mm	100 mm
OD	16 mm	20 mm
Wall	1.5 mm	1.5 mm
Hand f ₁	174.7 Hz FAIL	330.1 Hz PASS
Crash stress	68.1 MPa	34.3 MPa
Crash yield SF	4.05	8.05
Mast mass	22.1 g	23.5 g

Shortening raises stiffness with the inverse cube of length; increasing OD raises the section moment faster than it raises effective modal mass. Together the changes raise stiffness 3.58x for a 1.4 g mass penalty while improving strength.

9. Static FEA, Modal Validation, and Convergence

The revised idealized tube is meshed with 28,985 quadratic C3D10 tetrahedra (58,232 nodes, target size 1.2 mm) and solved in CalculiX. The root is fixed. The 128.8 N crash load is distributed over 474 tip-ring nodes to avoid a nonphysical point-load singularity. Stress is compared on a fixed mid-span gauge region; the global tip displacement provides the second static check.

Hand-to-FEA agreement

Quantity	Hand calc	FEA	Difference	+/-15%?
First bending frequency	330.1 Hz	285.5 Hz	-13.5%	PASS
Tip deflection	0.166 mm	0.176 mm	+5.9%	PASS
Mid-span gauge stress	17.1 MPa	17.4 MPa	+1.3%	PASS

The 41.9 MPa global peak occurs at the fixed-root re-entrant corner. It is a stress-concentration/mesh-singularity value and is not compared with elementary beam theory. The fixed gauge region and global deflection are the reportable static metrics.

Modal decision

The first two FEA modes are the near-degenerate bending pair at **285.5 Hz** and 301.3 Hz; the third is 1009.3 Hz. The higher-fidelity first mode remains above the 200 Hz guard by 1.43x and above the 100 Hz control rate by 2.86x. The roughly 14% reduction from the Rayleigh estimate is consistent with the closed-form model's rigid-root and Euler-Bernoulli assumptions.

Mesh convergence

Mesh	Gauge stress	Tip deflection	
3.0 mm / 4,747 elements	17.38 MPa	0.1772 mm	290.6 Hz
2.0 mm / 10,435 elements	17.50 MPa	0.1751 mm	289.6 Hz
1.2 mm / 28,985 elements	17.48 MPa	0.1757 mm	285.5 Hz
Final change	-0.15%	0.30%	-1.42%

All final changes are below 1.5%, an order of magnitude inside the 5% gate. Sources: `runs/mast_fea/fea_summary.txt` and `runs/mast_fea/mesh_convergence.txt`.

10. Tolerances, Installation, and the Physical Compliance Gate

Tolerance-driven design decision

The UST-10LX scans a horizontal plane, so the governing stack is pitch/roll tilt rather than software-calibratable yaw. At a 0.120 m assumed deck height, the optical center is 0.220 m above the floor. A 0.30 m wall at 10 m creates a 0.458 deg up-tilt bound.

The deterministic run at `runs/mast_tolerance_stack/summary.txt` reports:

Condition	Tilt	Verdict
Blind worst-case stack	1.284 deg	FAIL
Blind RSS stack	0.652 deg	FAIL
After scan-plane leveling	0.354 deg	PASS, 1.29x margin

The failure creates two requirements:

- 1 level the scan plane once at installation by shimming the mast base and equalizing wall return heights at two distances; and
- 1 keep the mast-root deck plane ≤ 0.138 m above the floor for the calibrated residual to clear the 0.30 m wall bound.

The tolerance inputs remain assumed until CAD and inspection replace them. The conclusion is still useful: blind assembly is unacceptable, and the design now has an install procedure plus a hard deck-height limit.

Physical Compliance Gate

Experiment registered - calibrated compliance campaign scheduled for July 25-August 2, 2026; measurement pending.

The frozen protocol tests the 100 mm-long, 20 mm-OD, 1.5 mm-wall 6061-T6 mast in both orthogonal axes:

- approximately 4, 8, 12, 16, and 20 N;
- at least three complete load/unload cycles at every level;
- 0.001 mm indicator resolution or better;
- fixture-motion subtraction on every row;
- force and displacement calibration; and
- saved raw rows `axis, cycle, direction, force_n, tip_mm, fixture_mm`.

The analyzer fits compliance with a free intercept. A result is INCONCLUSIVE if either axis has $R^2 < 0.99$, hysteresis $> 5\%$ of full-scale displacement, missing fixture motion, or relative expanded uncertainty $U95 > 10\%$. If quality passes, each measured as-built compliance must be within $\pm 15\%$ of the as-built FEA prediction for VALIDATED; otherwise the verdict is DISCREPANCY. Thresholds do not move after data is visible.

The experiment measures compliance only. It **does not validate stress**, fatigue life, on-car vibration, perception, or crash survival.

Source: <docs/specs/mast-physical-validation/design.md>.

11. Integration Risks, Limitations, and Next Gates

Known limitations

- The controls and identification results are simulation-only and use one map.
- Oracle recovery is not physical parameter validation.
- MPC timing is implementation-specific; its maximum solve time exceeds one control period.
- FEA uses an idealized tube and fixed root; clamp, fastener, and as-built geometry are not yet represented.
- The 50 g shock, bracket/cable mass, and several tolerance contributors are explicit assumptions.
- The 4.42 kg mass is a bottom-up estimate, not a scale measurement.
- No on-car excitation spectrum exists, so separation from operational vibration remains unvalidated.
- No physical mast or vehicle result is claimed.

Downstream gates

1 **CAD:** model the $L=100$ mm / $OD=20$ mm / $t=1.5$ mm mast and deck interface; enforce deck ≤ 0.138 m and the required front transponder bay.

1 **Fabrication:** issue a drawing/traveler, record stock certificate or seller specification, clamp/fastener condition, process, and lead time.

1 **Inspection:** record as-built length, OD, wall, clamp engagement, load height, and both test axes.

1 **As-built model:** regenerate hand and FEA compliance with those dimensions.

1 **Campaign:** execute the frozen matrix and publish the raw data, calibration, uncertainty, plot, and verdict.

1 **Hardware controls:** repeat excitation and held-out identification on the physical platform; do not transfer simulator coefficients.

The report is complete because each unresolved engineering activity has a named input, owner-facing action, and acceptance gate. It is not completed by silently converting pending evidence into a result.

12. Traceability and Reproduction

Mechanical proof-chain artifacts

Claim	Committed source
Baseline loads, strength, deflection, 174.7 Hz failure	runs/mast_hand_calc/summary.txt
Revised geometry, 330.1 Hz prediction, SF 8.05	runs/mast_hand_calc/design_sweep.txt
285.5 Hz / 0.176 mm / 17.4 MPa FEA comparison	runs/mast_fea/fea_summary.txt
-0.15% / 0.30% / -1.42% final mesh changes	runs/mast_fea/mesh_convergence.txt
1.284 deg fail / 0.354 deg pass / 0.138 m deck limit	runs/mast_tolerance_stack/summary.txt
4.42 kg vs 4.5 kg mass budget	docs/design/16_mechanical_design_analysis.md
Frozen physical +/-15% verdict	docs/specs/mast-physical-validation/design.md

Reproduction commands

```
# Deterministic hand model and redesign sweep
python experiments/mast_hand_calc.py

# Tolerance stack and committed summary
python experiments/mast_tolerance_stack.py

# Physical gate regression tests (measurement remains pending)
python experiments/test_mast_physical_validation.py

# Item-12 traceability and PDF acceptance
python experiments/test_final_report.py

# Build the review PDF
python scripts/build_final_report.py
```

The FEA setup and its two-environment gmsh/CalculiX commands are documented in [docs/design/FEA_SETUP.md](#). The checked-in summaries are the review evidence; large mesh and result binaries remain excluded from version control.

Closing statement

The design review now stands on a reproducible failure-and-redesign chain: traceable simulation envelope, closed-form calculation, a failed modal criterion, a targeted geometry change, model-to-model agreement inside the frozen band, converged FEA metrics, and a preregistered physical gate that remains visibly pending until accepted data exists.